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Mukesh Patel School of Technology Management & Engineering

Course Policy Document

Course Name - (Code): Image and Video Processing – 702EX0E020

Program and Semester: B Tech (EXTC/Computer Science/ CSE(DS) 311 (VT)/B Tech & MBA Tech (Information Technology/Computer Engineering)/BTI (Computer Engineering/ Information Technology) **Semester:** V/ VI/ VIII/ X/ XII	**Pre-requisite Course:** Basic operations on 2D arrays
Academic Year:2026-27	**Credit Details:** L T P C H / 2 0 2 3 4
Name of Course Faculty: Dr. Sachin Sonawane **Program:** B Tech Computer Science, B Tech / MBA Tech Computer Engineering **Contact Details:** sachin.sonawane@nmims.edu **Office Hours:** 10 am – 5 pm	**Faculty associated with the course: -** 1.
Pre-Course Activity: Revise the mathematical concepts, such as matrix operations, derivatives, integrals, and basic probability distributions along with programming skills	
Course link:	

1. Introduction to the Course

1.1 Importance of the Course

1.1.1 **Domain** Relevance**:**

Imageprocessing techniques can improve the visual quality of images, making them clearer, sharper, and more informative. It makes the tasks such as object recognition, pattern recognition and decision-making etceasy and accurate in real time applications.

1.1.2 **Industry** Relevance**:**

Image and Video Processing plays a crucial role in numerous Industrial sectors like medical imaging, Biometrics, Industry Automation, Media and entertainment, security and surveillance which impacting various aspects of modern life. Companies like Amazon, Google, Microsoft, Yahoo etc are looking for the engineering with basic image processing knowledge and good programming skills.

1.2 Objectives of the Course:

This course introduces concepts, methodologies and performance metrics for still image and motion picture processing. It also helps to develop a foundation for further study and research in the signal processing domain.

2. Course Outcomes (CO), Mapping with Program Outcomes (PO), and Program Specific Outcomes (PSO)

2.1 Course Outcomes

2.1.1 CO1: Explain the fundamentals of digital images, including distance measures, and file formats. (L2 - Understanding)

2.1.2 CO2: Apply image enhancement techniques such as point processing, spatial filtering, and histogram equalization for image improvement and image transforms for analyzing and compressing images. (L3 - Applying)

2.1.3 CO3: Analyze morphological image processing techniques and edge detection methods for feature extraction. (L4 - Analyzing)

2.1.4 CO4: Compare different image segmentation methods, including thresholding, region-based segmentation, and the Otsu method. (L4 - Analyzing)

2.1.5 CO5: Explain video representation concepts and motion estimation techniques used in digital video compression. (L2 - Understanding)

2.2 Program Outcomes (PO) the course contributed to:

2.2.1 PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.

2.2.2 PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

2.2.3 PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.

2.2.4 PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based

knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.

2.2.5 PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems

2.2.6 PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

2.2.7 PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.

2.2.8 PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

2.2.9 PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

2.2.10 PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

2.2.11 PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

2.3 Program Specific Outcomes (PSO):

2.3.1 PSO-1: Apply elemental principles to conceptualize, design, and implement innovative solutions for communication, networking, and signal processing applications.

2.3.2 PSO-2: Exhibit expertise and adaptability in developing efficient and pioneering technologies in Embedded Systems, Artificial Intelligence, and Next-Generation Communication systems.

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	-	-	-	-	-	-	-	-	2	3	-
CO2	3	3	3	-	3	-	-	-	-	-	2	3	3
CO3	3	3	-	3	-	-	-	-	-	-	2	3	2
CO4	3	3	-	3	-	-	-	-	-	-	2	3	2
CO5	3	2	-	-	-	-	-	-	-	-	2	3	2

Mapping Levels: 1- High, 2-Medium, 3-Low

2.4 Student Outcomes (SO) (For ABET accredited Programs):

2.4.1 **SO-1:** An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

2.4.2 **SO-2:** An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

2.4.3 **SO-3:** An ability to communicate effectively with a range of audiences.

2.4.4 **SO-4:** An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.

2.4.5 **SO-5:** An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

2.4.6 **SO-6:** An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

2.4.7 **SO-7:** An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

3. Teaching-learning methodology

3.1Instruction Plan

Lecture No.	Topic	Teaching Method *	Blooms Level	Resources	COs mapped	Assessment and Evaluation
Unit 1:Image fundamentals						
1	Representation of digital image, image as a matrix, image as a function, basics of colour image, spatial and gray level resolution,	PPT, Whiteboard	Understand	Pages: 65-68 NPTEL Course	CO1	CT1,Lab Performance, Group Activity and TEE
2	Distance measures-Euclidean distance and m path, image file formats (TIFF, PNG, JPEG).	PPT, Whiteboard	Understand	Pages: 61-64, 69-78 Pages: 79-83	CO1	CT1,Lab Performance, Group Activity and TEE
Unit 2: Image enhancement						
3	Point processing techniques: Linear- digital negative, Piecewise linear-contrast stretching,	PPT, Whiteboard	Apply	Pages: 120-121	CO2	CT1,Lab Performance, Group Activity and TEE
4	Thresholding, Intensity level slicing	PPT, Whiteboard	Apply		CO2	CT1,Lab Performance, Group Activity and TEE
5	Nonlinear- log transformation, power law transformation	PPT, Whiteboard	Apply	Pages: 122-132	CO2	CT1,Lab, Performance, Group Activity and TEE
6	Neighborhood processing techniques: smoothing spatial filters,	PPT, Whiteboard	Apply	Pages 153-174	CO2	CT1,Lab, Performance, Group Activity and TEE

7	Sharpening spatial filters, median filters, Gaussian filtering.	PPT, Whiteboard	Apply	Pages 175-187	CO2	CT1,Lab, Performance, Group Activity and TEE
8	Histogram processing-histogram equalization, Image in Frequency domain: Fourier spectrum, Fourier analysis, filtering in frequency domain	PPT, Whiteboard	Apply	Pages: 133-152	CO2	CT1,Lab, Performance, Group Activity and TEE
Unit 3: Image transforms						
9	Image transforms Discrete Fourier Transform, Hadamard transform	PPT, Whiteboard	Apply	Jain Textbook Pages: 155-158	CO2	CT1,Lab, Performance, Group Activity and TEE
10	Discrete cosine transform and its applications for image compression	PPT, Whiteboard	Apply	Pages: 487-490 Jain Textbook Pages: 150-153	CO2	CT1,Lab, Performance, Group Activity and TEE
Unit 4: Morphological image processing						
11	Dilation, erosion, Opening, Closing	PPT, Whiteboard	Analyze	Pages: 635-650	CO3	CT2,Lab Performance, Group Activity and TEE
12	Hit or-Miss transformation, Basic morphological algorithms-boundary extraction on binary images	PPT, Whiteboard	Analyze	Pages:652-660	CO3	CT2,Lab Performance, Group Activity and TEE
13	Thinning, thickening	PPT, Whiteboard	Analyze	Pages: 660-662	CO3	CT2,Lab Performance, Group Activity and TEE

14	Gray scale morphology	PPT, Whiteboard	Analyze	Pages: 674-679	CO3	CT2,Lab Performance, Group Activity and TEE
Unit 5: Edge Detection						
15	Gradient based edge detection-First order and second order derivative	PPT, Whiteboard	Analyze	Pages:699-724	CO3	CT2,Lab Performance, Group Activity and TEE
16	Edge detection operators- Prewitt, Sobel, Laplacian, Laplacian of Gaussian, canny edge detector.	PPT, Whiteboard	Analyze		CO3	CT2,Lab Performance, Group Activity and TEE
Unit 6: Image segmentation						
17	Basic relationships between pixels, 4, 8 and m connectivity, Segmentation using thresholding	PPT, Whiteboard	Analyze	Pages: 79-83	CO4	CT2,Lab Performance, Group Activity and TEE
18	Bayesian classification of foreground and background pixels, expectation- maximization theorem for binarization, Otsu method, Otsu method	PPT, Whiteboard	Analyze	Pages: 742-751	CO4	CT2,Lab Performance, Group Activity and TEE
19	Region based segmentation, Region growing	PPT, Whiteboard	Analyze	Pages: 764-770	CO4	CT2,Lab Performance, Group Activity and TEE
20	Region splitting and merging	PPT, Whiteboard	Analyze		CO4	CT2,Lab Performance, Group Activity and TEE
Unit 7: Colour Image Processing						
21	Colour fundamentals and colour models (RGB)	PPT, Whiteboard	Understand	Pages: 400-408	CO1	Group Activity and TEE
22	Colour models (CMY, CMYK, HSI, YUV, HSV)	PPT, Whiteboard	Understand	Pages: 408-413	CO1	Group Activity and TEE

23	Model transformation, colour slicing	PPT, Whiteboard	Apply	Pages: 430-437	CO2	Group Activity and TEE
24	Histogram processing of colour images	PPT, Whiteboard	Apply	Pages: 439-441	CO2	Group Activity and TEE
Unit 8: Fundamentals of digital video and its compression						
25	Video representation-temporal correlation	PPT, Whiteboard	Understand	Image and video processing handbook Pages:	CO5	Lab Performance, Group Activity and TEE
26	Video frame classifications, I, P and B frames	PPT, Whiteboard	Understand	Image and video processing handbook	CO5	Lab Performance, Group Activity and TEE
27	Fundamentals of motion estimation	PPT, Whiteboard	Understand	Image and video processing handbook	CO5	Group Activity and TEE
28	Motion estimation algorithms-exhaustive search block matching	PPT, Whiteboard	Understand	Image and video processing handbook	CO5	Group Activity and TEE
29	2D-log search method	PPT, Whiteboard	Understand	Image and video processing handbook	CO5	Assignment, Group Activity and TEE

30	3 step search method	PPT, Whiteboard	Understand	 Image and video processing handbook	CO5	Assignment, Group Activity and TEE

4. Assessment and Evaluation Scheme

	Internal Continuous Assessment (ICA) (50 Marks)					
	Class Tests (20 Marks)		**Term-work (30 Marks)**			
Assessment Component	Class Test-1	Class Test-2	Lab Performance	Mini Project	Research Paper/Report	Class Participation/viva/Quiz
Marks	10	10	10	10	5	5

4.1 Internal Continuous Assessment (ICA) – 50 marks

- **Class Test-1 (10 Marks)**
  - To be conducted in the 6th week of the semester.
  - Duration: 45 minutes.
  - Test will include short and descriptive questions based on Units 1, 2 and 3 of the syllabus.
  - Evaluation will be based on accuracy and conceptual clarity

- **Class Test-2 (10 Marks)**
  - To be conducted in the 11th week of the semester.
  - Duration: 45 minutes.
  - Content will cover Units 4, 5 and 6 of the syllabus.
  - Similar to Class Test-1, the test will emphasize problem-solving and analytical skills.

- **Lab Performance (10 Marks)**

  Evaluation will be based on:
  - Active participation in practicals and demonstration of the understanding and application of concepts in lab sessions (5 marks)
  - Timely submission of weekly programming assignments (5 marks)

- **Mini Project (10 Marks)**

  Students (in group of 3-4) are required to undertake a mini project based on topics related to experiments in the syllabus.

  Evaluation Criteria:

- Timely submission (3marks)
- Working and completeness of the project (4marks)
- Quality of documentation (3marks)

- **Research Paper/Report (5 Marks)**
  Students (in group of 3-4)are required to prepare a research paper on a topic related to Image and Video Processing.
  The paper should preferably follow the IEEE format and include:Abstract, Introduction, Methodology, Results or discussion (based on simulation/analysis/literature review), Conclusion, References
  Evaluation Criteria:
  - Relevance and originality of the topic
  - Technical content and depth of understanding
  - Quality of writing, formatting, and references
- **Class Participation (5 Marks)**

4.2 Term End Examination (TEE) – (100 marks scaled down to 50)

TEE conducted at the end of the semester will be for 100 marks (3-hour duration). Marks obtained scaled down to 50. There will be 7 questions (each question will be 20 marks). Q1 will be compulsory and any 4 from the remaining to be solved.

4.3 Course Passing Criteria

4.3.1. ICA (50 marks) – No minimum marks

4.3.2. TEE (100 marks scaled to 50) – 40% required for passing

4.3.3. (ICA + TEE) (100 marks) – 40%required for passing

4.4 Assessments and Mapping to Course Outcomes

	Internal Continuous Assessment (ICA)						**Term End Examination (TEE)**
Course Outcomes	CT-1	CT-2	Lab Performance	Mini Project	Research Paper/Report	Class Participation	TEE
CO-1	Y		Y	Y	Y		Y
CO-2	Y		Y	Y	Y		Y
CO-3		Y	Y	Y	Y		Y
CO-4		Y	Y	Y	Y		Y
CO-5			Y	Y	Y		Y

5. Laboratory details

Knowledge of Python/MATLAB for laboratory exercise is a prerequisite. Students are expected to recall the fundamental theory concepts relevant to the exercise to be perform in the upcoming laboratory. The following 10 programming exercises and Group activity on the application of image/video processing techniques beyond syllabus (to be done by the students) will form the submission for laboratory coursework.

Exp. No.	**Week No.#**	**Experiment Topics**	**Mapped CO**
1.	1	Apply image processing algorithms on the test image to obtain new image. i. Open, show and save images using different techniques. ii. To observe the effect of different gray level images on their corresponding image matrix. iii. Image conversion (Color to Gray, Gray to Binary, Color to Binary). iv. Covert image formats. Comment on subjective quality of the image for obtained output.	CO1
2.	2	Apply distance measures in the given image	CO2
3.	3	For the given test image, apply point processing operations Comment on the comparison and performance of both the techniques.	CO2
4.	4	Apply histogram equalization and spatial domain filter techniques	CO2
	5	Unit 1 Week	
5.	6	Apply Edge detection operations on the image.	CO2
6.	7	Morphological Operations	CO3
7.	8	Frequency domain filter, DFT, DCT Transform	CO3
	9	Unit 2 Week	CO2
8.	10	Local and Adaptive Thresholding, OTSU Segmentation	
9.	11	Color Image Processing	CO4
10.	12	Video Processing	CO5
	13	Evaluation of Group activity Phase	
	14,15	Final submission of Group activity task	

6. Tutorial Plan

This course does not have any tutorial

7. Course Material

References and Lab Manuals would be uploaded on LMS by faculty every week.

NPTEL Video Lecturs
Link:https://www.youtube.com/watch?v=CVV0TvNK6pk&list=PL1F076D1A98071E24

8. GenAI Usage

8.1 Pre-class Activity:

- Identify any one Digital Image processing application and relate the concepts taught in the previous class.
- How machine learning and deep learning techniques can improve the accuracy of the image processing application.
- Study the advanced edge detection techniques and compare with the traditional edge detection techniques.

8.2 In-Class Activity:

- Understand the role of enhancement, edge detection and compression techniques in real time applications.

8.3 Assignments

GenAI is not to be used for doing Assignments given by faculty.

8.4 Group Activity

GenAI is permitted to be used. Students need to cite the references used.

9. Academic Integrity Statement

Original work expected from students for all of the **assigned assessment work**. Copying in any form not acceptable and will invite strict disciplinary action. Evaluation of corresponding component will be affected proportionately in such cases. Plagiarism detection software will be used to check plagiarism wherever applicable. Academic integrity is expected from students in all components of course assessment**.**

*** - Only Teaching Method in the Instruction Plan for the course may vary for different faculty teaching the course, rest of the Course Policy Document will not change.**